DevOps

Kubernetes Dashboard

ISTAD

Kubernetes Dashboard

```yaml
apiVersion: cert-manager.io/v1
kind: ClusterIssuer
metadata:
  name: all-clusterissuer
spec:
     acme:
       server: https://acme-v02.api.letsencrypt.org/directory
       email: leangsengk90@gmail.com
       privateKeySecretRef:
         name: letsencrypt-key
       solvers:
         - http01:
             ingress:
               class: nginx
```

Kubernetes Dashboard

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
     name: kubernetes-ingress
     namespace: kube-system
     annotations:
        cert-manager.io/cluster-issuer: all-clusterissuer
        kubernetes.io/ingress.class: nginx
        nginx.ingress.kubernetes.io/backend-protocol: "HTTPS”
spec:
     tls:
       - hosts:
           - dashboard.e-crops.co
         secretName: kubernetes-tls
     rules:
     - host: dashboard.e-crops.co
       http:
         paths:
         - pathType: Prefix
           path: "/"
           backend:
             service:
               name: kubernetes-dashboard
               port:
                 number: 443
```

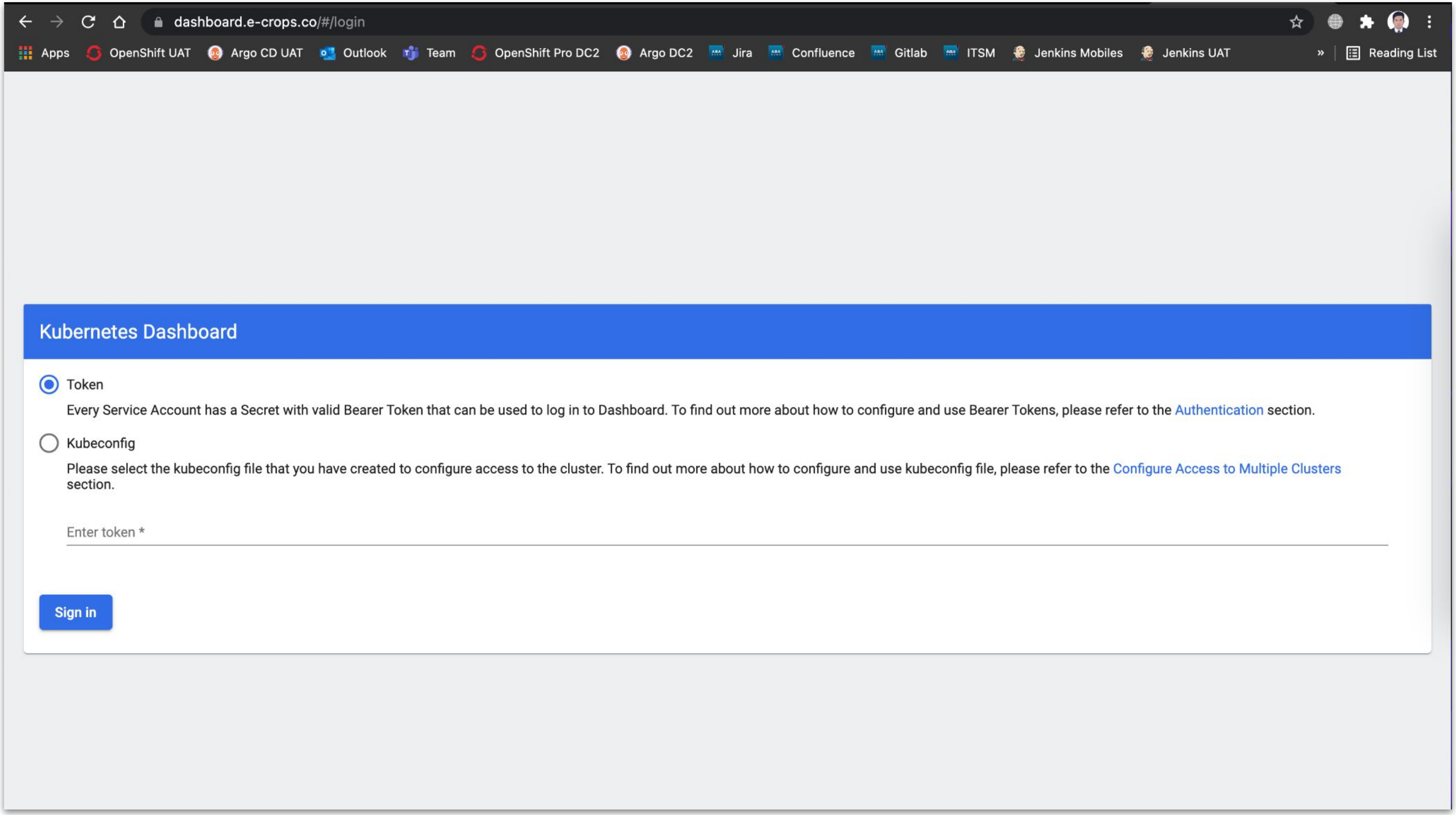
ISTAD

Kubernetes Dashboard

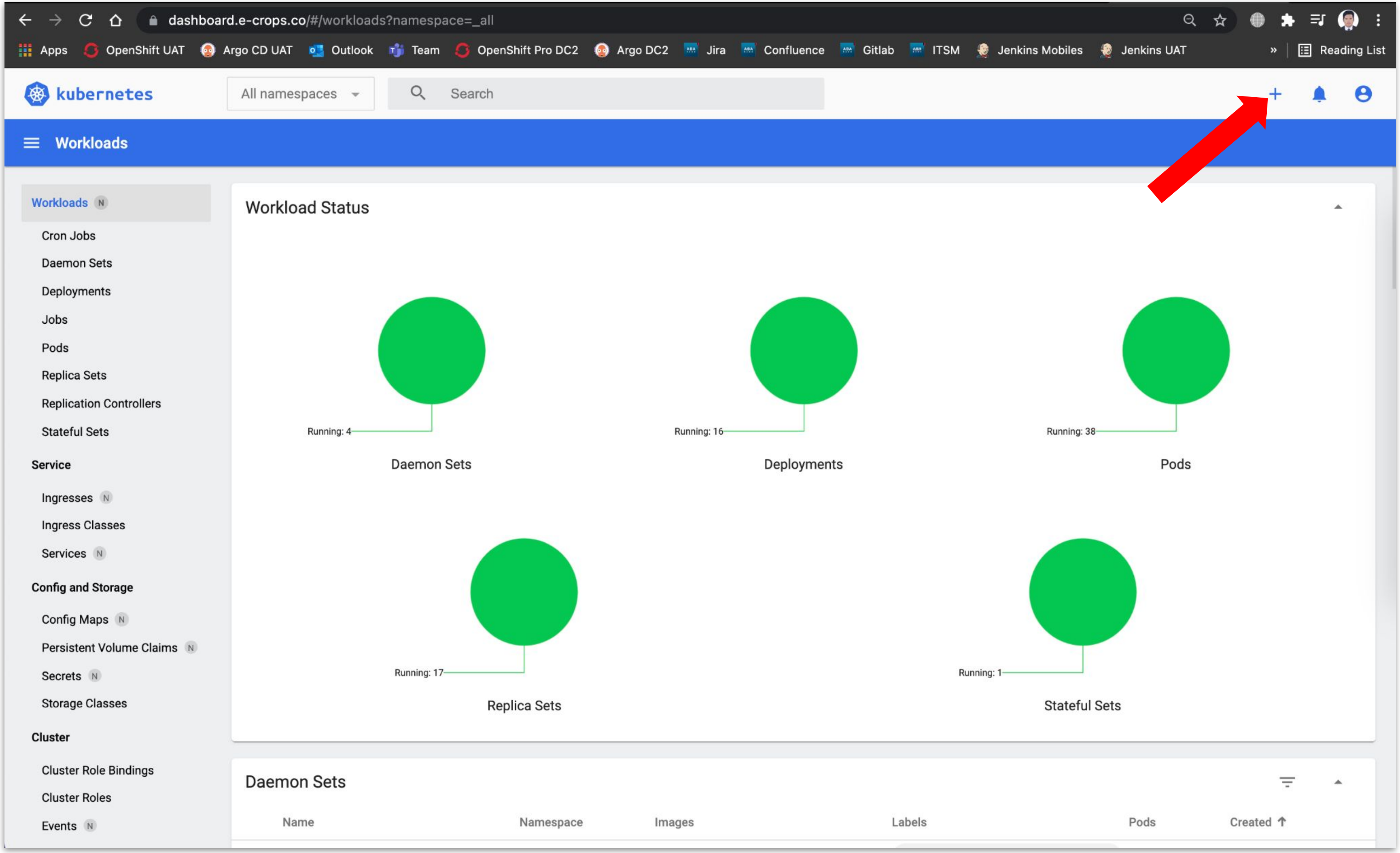
```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: kubernetes-dashboard
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: admin-user
  namespace: kubernetes-dashboard
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: admin-user
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: cluster-admin
subjects:
- kind: ServiceAccount
  name: admin-user
  namespace: kubernetes-dashboard
```

Kubernetes Dashboard

- `kubectl -n kubernetes-dashboard create token admin-user --duration=24h`

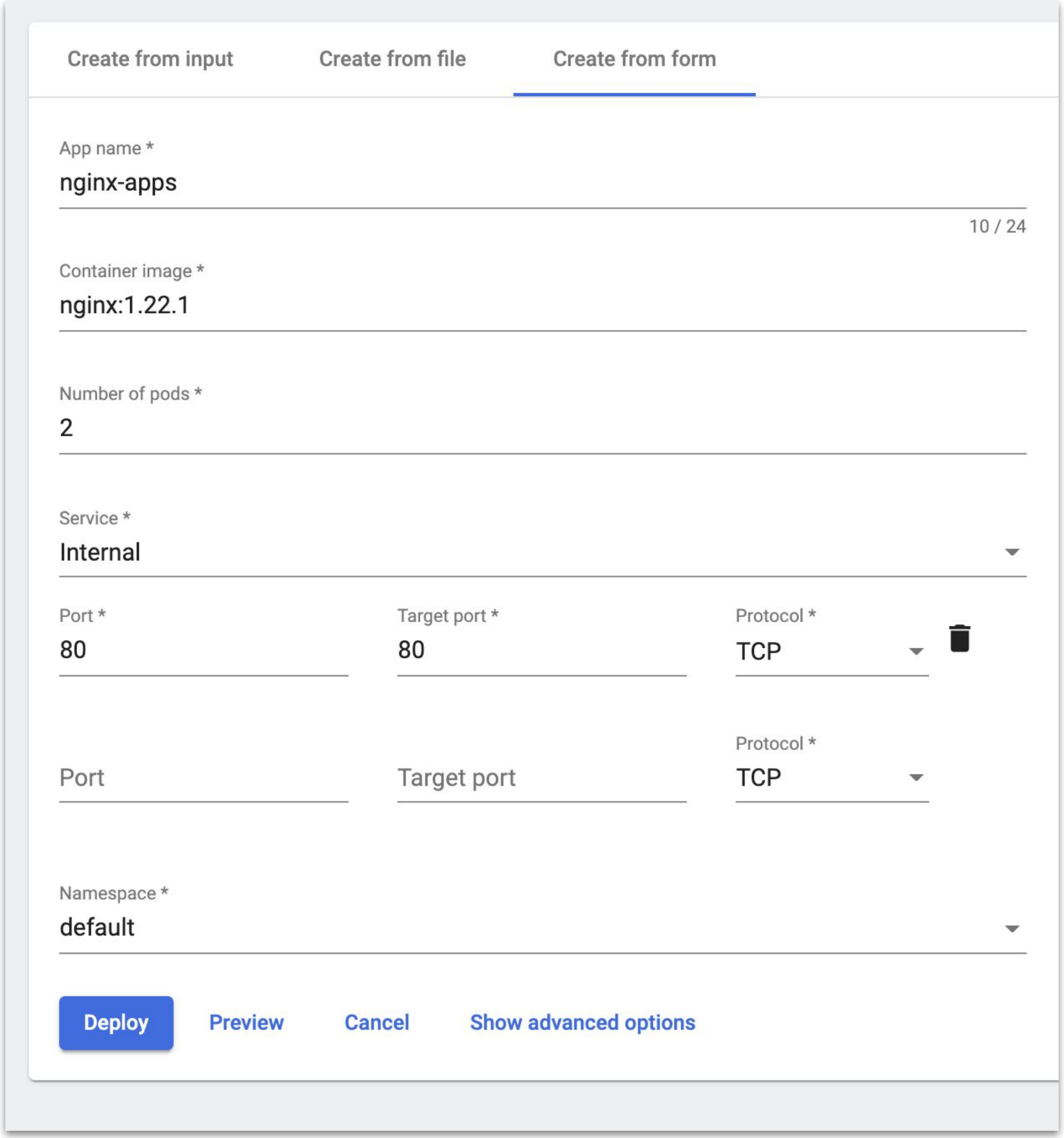


Create Kubernetes Objects



iSTAD

Create Deployment and Service from form



ISTAD

Create Ingress from input

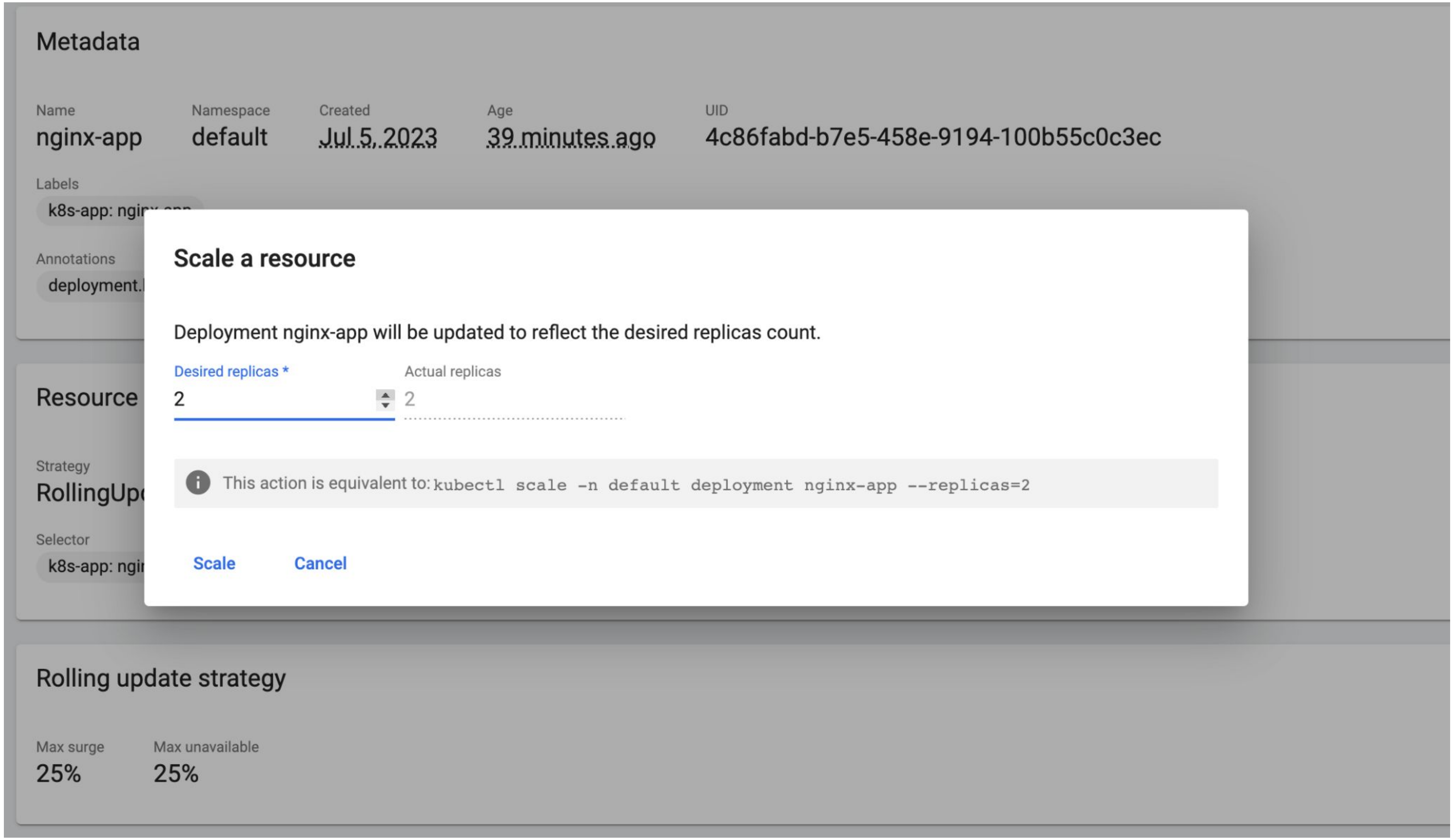
Create from input | Create from file | Create from form

Enter YAML or JSON content specifying the resources to create to the currently selected namespace. Learn more

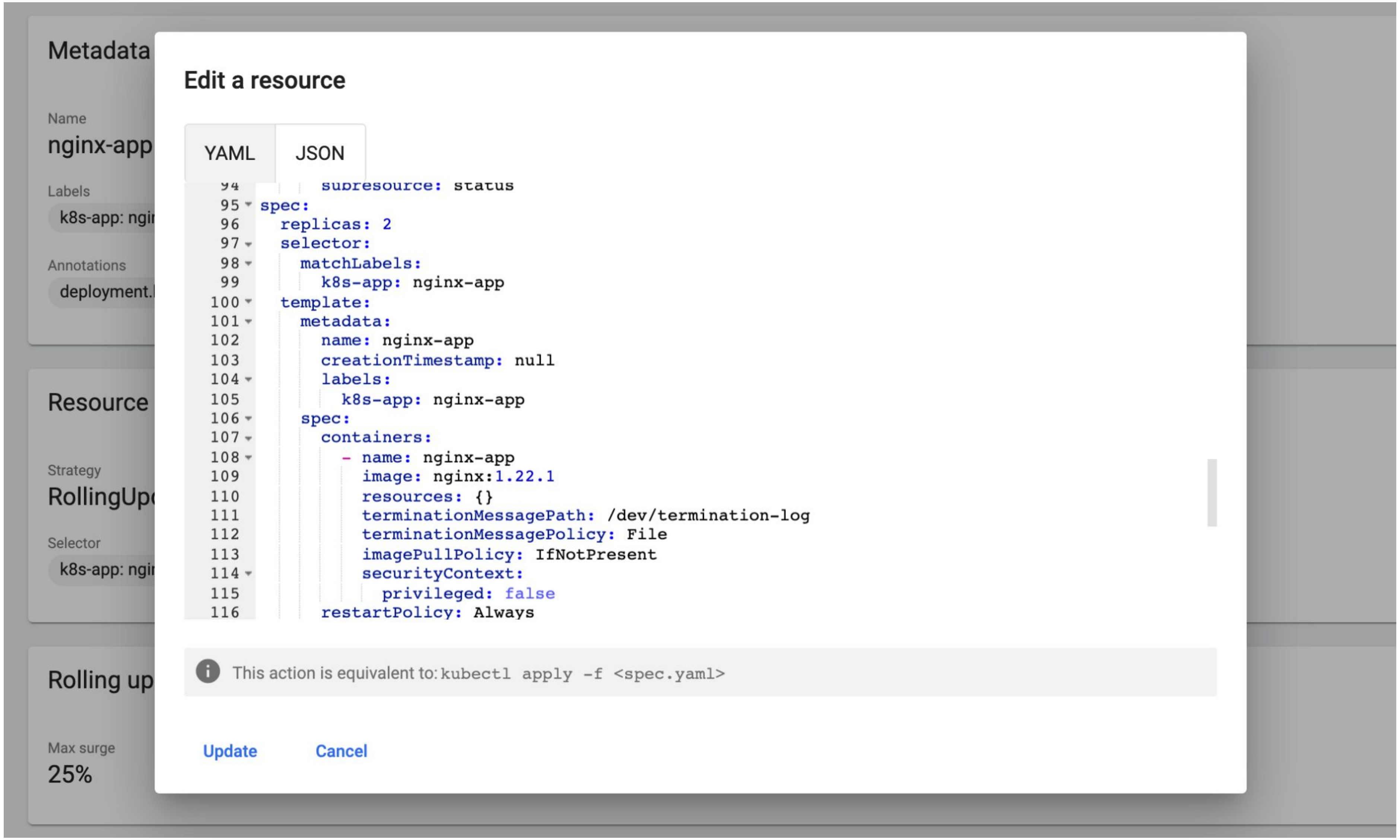
```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
  annotations:
    cert-manager.io/cluster-issuer: all-clusterissuer
    kubernetes.io/ingress.class: nginx
spec:
  tls:
    - hosts:
        - nginx.e-crops.co
      secretName: nginx-tls
  rules:
  - host: nginx.e-crops.co
    http:
      paths:
      - pathType: Prefix
        path: "/"
        backend:
          service:
            name: nginx-app
            port:
              number: 80
```

Upload   Cancel

Modify Kubernetes Objects - Scales

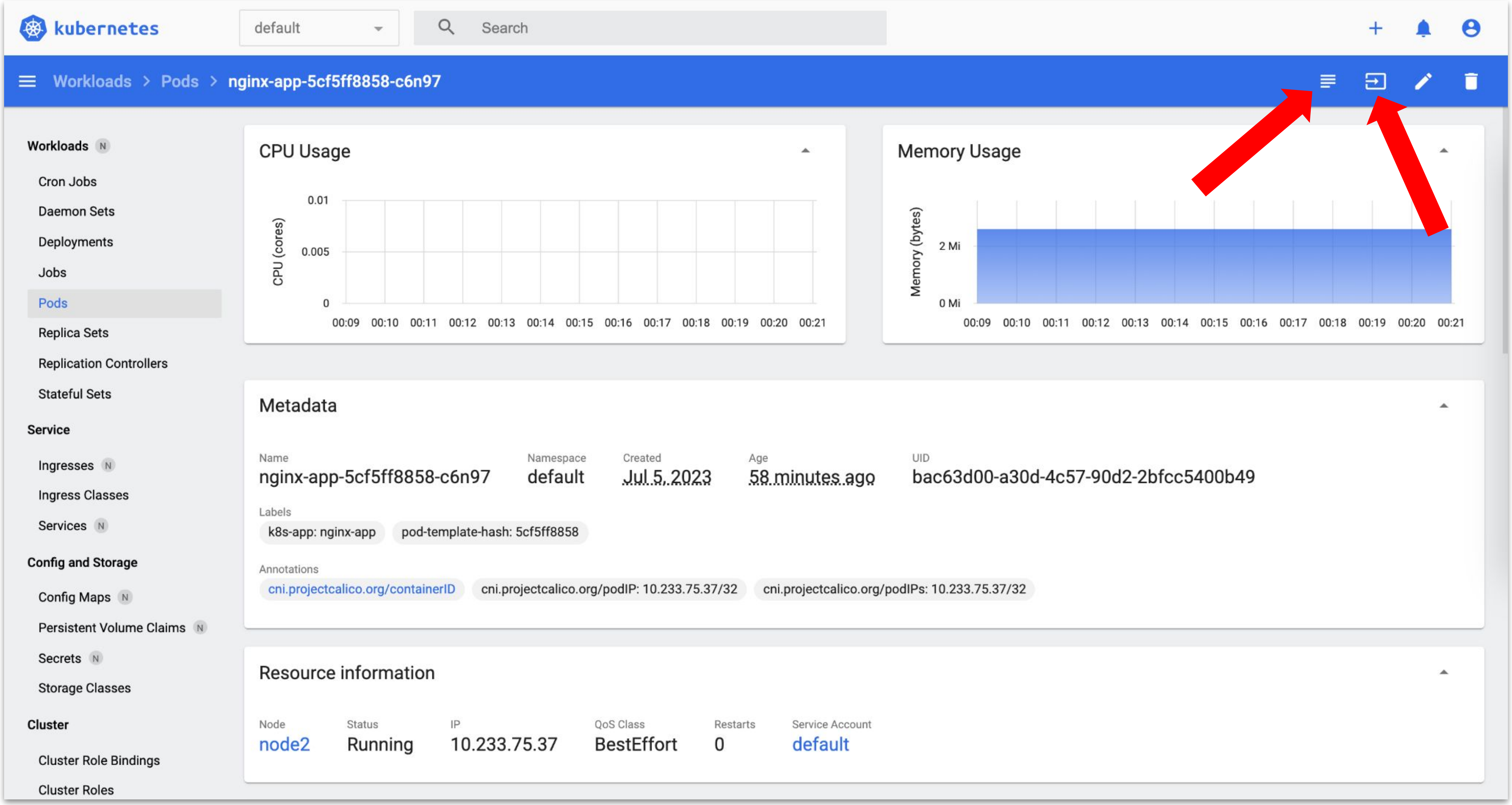


Modify Kubernetes Objects - Yaml

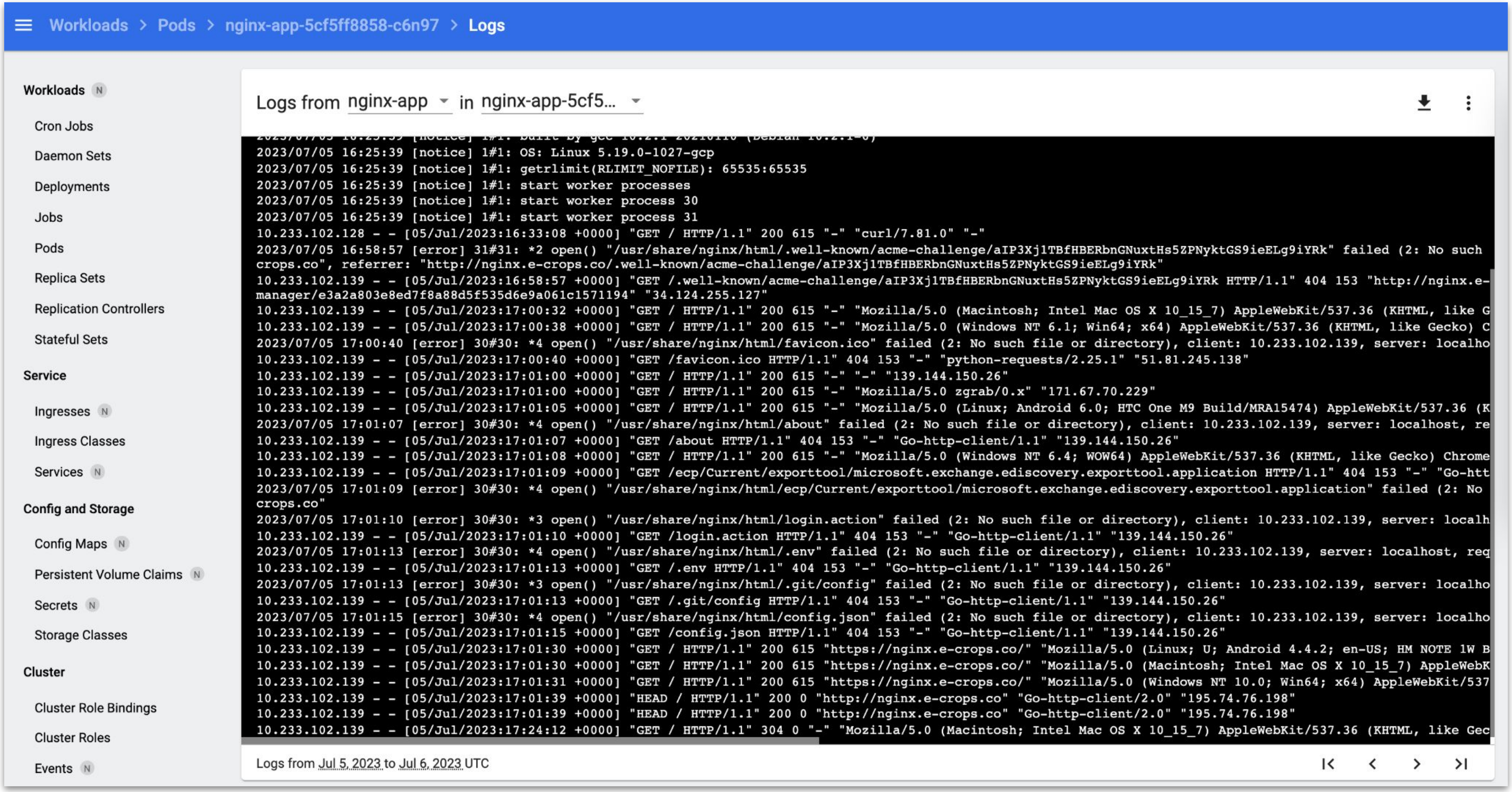


iSTAD

Deployment / Replica Set / Pod

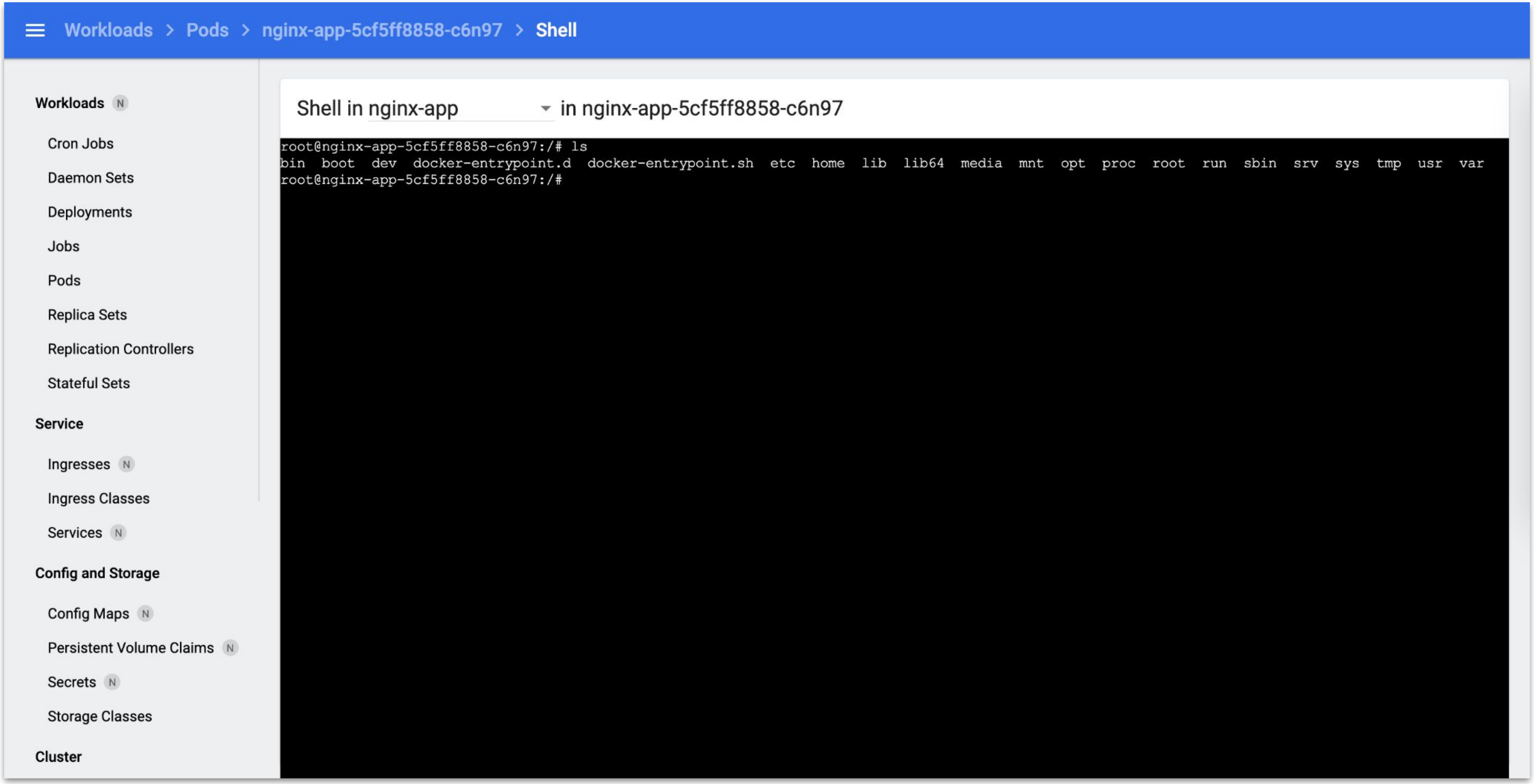


View Pod Log



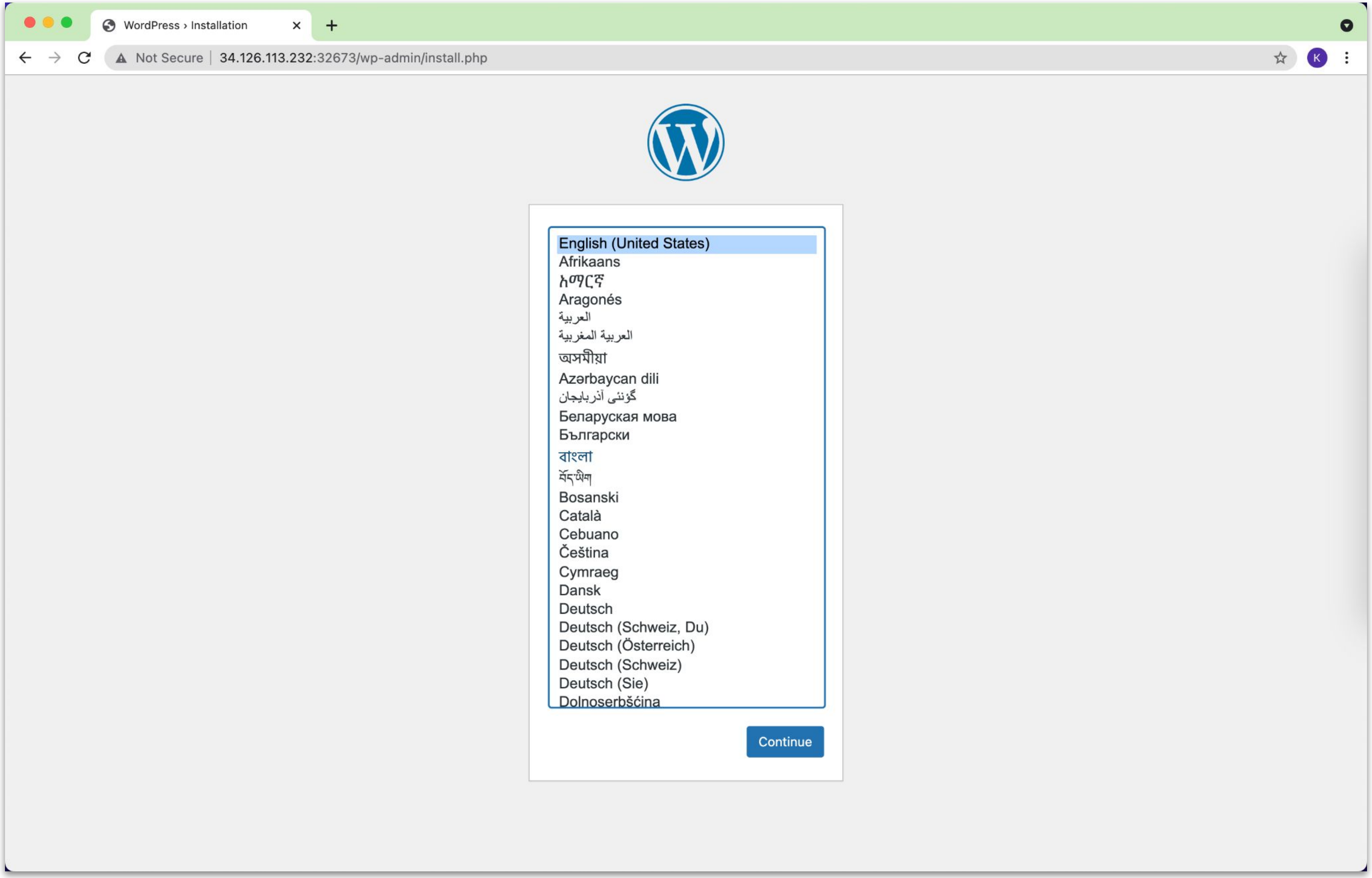
Workloads > Pods > nginx-app-5cf5ff8858-c6n97 > Logs
Workloads N
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses N
Ingress Classes
Services N
Config and Storage
Config Maps N
Persistent Volume Claims N
Secrets N
Storage Classes
Cluster
Cluster Role Bindings
Cluster Roles
Events N
Logs from nginx-app in nginx-app-5cf5...
2023/07/05 16:25:39 [notice] 1#1: OS: Linux 5.19.0-1027-gcp
2023/07/05 16:25:39 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 65535:65535
2023/07/05 16:25:39 [notice] 1#1: start worker processes
2023/07/05 16:25:39 [notice] 1#1: start worker process 30
2023/07/05 16:25:39 [notice] 1#1: start worker process 31
10.233.102.128 - - [05/Jul/2023:16:33:08 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/7.81.0" "-"
2023/07/05 16:58:57 [error] 31#31: *2 open() "/usr/share/nginx/html/.well-known/acme-challenge/aIP3Xj1TBfHBERbnGNuxtHs5ZPNyktGS9ieELg9iYRk" failed (2: No such
crops.co", referrer: "http://nginx.e-crops.co/.well-known/acme-challenge/aIP3Xj1TBfHBERbnGNuxtHs5ZPNyktGS9ieELg9iYRk"
10.233.102.139 - - [05/Jul/2023:16:58:57 +0000] "GET /.well-known/acme-challenge/aIP3Xj1TBfHBERbnGNuxtHs5ZPNyktGS9ieELg9iYRk HTTP/1.1" 404 153 "http://nginx.e-
manager/e3a2a803e8ed7f8a88d5f535d6e9a061c1571194" "34.124.255.127"
10.233.102.139 - - [05/Jul/2023:17:00:32 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like G
10.233.102.139 - - [05/Jul/2023:17:00:38 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) C
2023/07/05 17:00:40 [error] 30#30: *4 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 10.233.102.139, server: localho
10.233.102.139 - - [05/Jul/2023:17:00:40 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "-" "python-requests/2.25.1" "51.81.245.138"
10.233.102.139 - - [05/Jul/2023:17:01:00 +0000] "GET / HTTP/1.1" 200 615 "-" "-" "139.144.150.26"
10.233.102.139 - - [05/Jul/2023:17:01:00 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 zgrab/0.x" "171.67.70.229"
10.233.102.139 - - [05/Jul/2023:17:01:05 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Linux; Android 6.0; HTC One M9 Build/MRA15474) AppleWebKit/537.36 (K
2023/07/05 17:01:07 [error] 30#30: *4 open() "/usr/share/nginx/html/about" failed (2: No such file or directory), client: 10.233.102.139, server: localhost, re
10.233.102.139 - - [05/Jul/2023:17:01:07 +0000] "GET /about HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
10.233.102.139 - - [05/Jul/2023:17:01:08 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 6.4; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome
10.233.102.139 - - [05/Jul/2023:17:01:09 +0000] "GET /ecp/Current/exporttool/microsoft.exchange.ediscovery.exporttool.application HTTP/1.1" 404 153 "-" "Go-htt
2023/07/05 17:01:09 [error] 30#30: *4 open() "/usr/share/nginx/html/ecp/Current/exporttool/microsoft.exchange.ediscovery.exporttool.application" failed (2: No
crops.co"
2023/07/05 17:01:10 [error] 30#30: *3 open() "/usr/share/nginx/html/login.action" failed (2: No such file or directory), client: 10.233.102.139, server: localh
10.233.102.139 - - [05/Jul/2023:17:01:10 +0000] "GET /login.action HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
2023/07/05 17:01:13 [error] 30#30: *4 open() "/usr/share/nginx/html/.env" failed (2: No such file or directory), client: 10.233.102.139, server: localhost, req
10.233.102.139 - - [05/Jul/2023:17:01:13 +0000] "GET /.env HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
2023/07/05 17:01:13 [error] 30#30: *3 open() "/usr/share/nginx/html/.git/config" failed (2: No such file or directory), client: 10.233.102.139, server: localho
10.233.102.139 - - [05/Jul/2023:17:01:13 +0000] "GET /.git/config HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
2023/07/05 17:01:15 [error] 30#30: *4 open() "/usr/share/nginx/html/config.json" failed (2: No such file or directory), client: 10.233.102.139, server: localho
10.233.102.139 - - [05/Jul/2023:17:01:15 +0000] "GET /config.json HTTP/1.1" 404 153 "-" "Go-http-client/1.1" "139.144.150.26"
10.233.102.139 - - [05/Jul/2023:17:01:30 +0000] "GET / HTTP/1.1" 200 615 "https://nginx.e-crops.co/" "Mozilla/5.0 (Linux; U; Android 4.4.2; en-US; HM NOTE 1W B
10.233.102.139 - - [05/Jul/2023:17:01:30 +0000] "GET / HTTP/1.1" 200 615 "https://nginx.e-crops.co/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebK
10.233.102.139 - - [05/Jul/2023:17:01:31 +0000] "GET / HTTP/1.1" 200 615 "https://nginx.e-crops.co/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537
10.233.102.139 - - [05/Jul/2023:17:01:39 +0000] "HEAD / HTTP/1.1" 200 0 "http://nginx.e-crops.co" "Go-http-client/2.0" "195.74.76.198"
10.233.102.139 - - [05/Jul/2023:17:01:39 +0000] "HEAD / HTTP/1.1" 200 0 "http://nginx.e-crops.co" "Go-http-client/2.0" "195.74.76.198"
10.233.102.139 - - [05/Jul/2023:17:24:12 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gec
Logs from Jul 5, 2023 to Jul 6, 2023 UTC

Execute into Pod



Practice

- Create MySql and Wordpress for both Deployments and Services via Kubernetes Dashboard
- Resource: https://github.com/leangsengk90/ingress-exercise.git



Thank you

Begin with the theory, Start with the practical